

Chandrayaan-1, India's first mission to Moon, was launched successfully on October 22, 2008 from SDSC SHAR, Sriharikota. The spacecraft orbited around the Moon at a height of 100 km from the lunar surface and provided chemical, mineralogical and photo-geologic mapping of the Moon. The spacecraft carried a total of 11 scientific instruments developed in India and other countries.

After completion of the major mission objectives, the orbit was raised to 200 km in May 2009. The satellite made more than 3400 orbits around the moon in its life time. The mission was concluded after the communication with the spacecraft was lost on August 29, 2009.

The payloads onboard Chandrayaan-1 are,

- a. Terrain Mapping Camera (TMC)
- b. Hyper Spectral Imager (HySI)
- c. Lunar Laser Ranging Instrument (LLRI)
- d. High Energy X - ray Spectrometer (HEX)
- e. Moon Impact Probe(MIP)
- f. Chandrayaan-1 X-ray Spectrometer (C1XS)
- g. Near Infrared Spectrometer (SIR – 2)
- h. Sub keV Atom Reflecting Analyzer (SARA)
- i. Miniature Synthetic Aperture Radar (Mini SAR)
- j. Moon Mineralogy Mapper (M3)
- k. Radiation Dose Monitor (RADOM)

a. Terrain Mapping Camera (TMC)

The aim of TMC was to map topography of both near and far side of the Moon and prepare a 3-dimensional atlas with a high spatial and elevation resolution. TMC imaged Moon in the panchromatic spectral region of 0.5 to 0.85 μm , with a spatial/ ground resolution of 5 m and swath coverage of 20 km. TMC measured the solar radiation reflected / scattered from the Moon's surface. The dynamic range of the reflected signal was found to be quite large, represented by the two extreme targets – fresh crust rocks and mature mare soil.

TMC payload is developed by SAC, Ahmedabad, ISRO.



b. Hyper Spectral Imager (HySI)

The objective of HySI is to obtain spectroscopic data for mineralogical mapping of the lunar surface. The data from this instrument helped in improving the available information on mineral composition of the surface of Moon. Also, the study of data in deep crater regions/central peaks, which represents lower crust or upper mantle material, may help in understanding the mineralogical composition of Moon's interior.



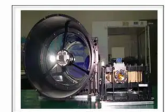
The uniqueness of the HySI was in its capability of mapping the lunar surface in 64 contiguous bands in the VNIR, the spectral range of 0.4-0.95 μm region with a spectral resolution of better than 15 nm and spatial resolution of 80 m, with swath coverage of 20 km. HySI collected the

Sun's reflected light from the Moon's surface through a tele-centric refractive optics and focus on to an APS area detector for this purpose.

The dispersion was achieved by using a wedge filter so as to reduce the weight and compactness of the system compared to using a prism / grating. The wedge filter was an interference filter with varying thickness along one dimension so that the transmitted spectral range varies in that direction. The wedge filter was placed in close proximity to an area detector. Thus, different pixels in a row of the detector received irradiance from the same spectral region but different spatial regions in the across track direction. In the column direction of the detector, different rows received irradiance of different spectral as well as spatial regions in the along track direction. The full spectrum of a target was obtained by acquiring image data in push broom mode, as the satellite moves along the column direction of the detector. An Active Pixel Sensor (APS) area array detector with built-in digitizer mapped the spectral bands. The payload mass was 2.5 kg and its size is 275 mm x 255 mm x 205 mm. HySI payload is developed by SAC, ISRO

c. Lunar Laser Ranging Instrument (LLRI)

The objective of LLRI was to provide ranging data for determining the height difference between the spacecraft and the lunar surface. LLRI worked on the time-Of-Flight (TOF) principle. In this method, a coherent pulse of light from a high power laser is directed towards the target whose range is to be measured. A fraction of the light is scattered back in the direction of the laser source where an optical receiver collects it and focuses it on to a photoelectric detector.



By accurately measuring the roundtrip travel time of the laser pulse, highly accurate range/spot elevation measurements can be made. LLRI consisted of a 10 mJ Nd:YAG laser with 1064 nm wave source operating at 10 Hz pulse repetition mode. The reflected laser pulse from the lunar surface is collected by a 200 mm Ritchey-Chrétien Optical receiver and focused on to a Silicon Avalanche Photodetector. The output of the detector was amplified and threshold detected for generating range information to an accuracy < 5m. Four constant fraction discriminators provided the slope information in addition to range information. The different modes of operation of LLRI and the range computations from the detector output were controlled and computed by a FPGA based electronics. The processed outputs of LLRI was used for generating high accuracy lunar topography. The payload mass was 11.37 kg with base plate. LLRI payload is developed by LEOS, ISRO.

d. Energy X - ray Spectrometer (HEX)

The scientific objectives of HEX was to:

- I. Identify excess ^{210}Pb in lunar polar regions deposited there as a result of transport of gaseous ^{222}Rn , a decay product of ^{238}U from other regions of the Moon. This will enable us to understand transport of other volatiles such as water to the polar regions.
- II. Detect other radioactive emissions, to characterise various lunar terrains for their chemical and radioactive composition on the basis of specific/integrated signal in the 30-270 keV region.
- III. Explore the possibility of identifying polar regions covered by thick water-ice deposit from a study of the continuum background



HEX covers the hard X-ray region from 30 keV to 270 keV. The experiment is designed primarily to study the emission of low energy (30-270 keV) natural gamma-rays from the lunar surface due to ^{238}U and ^{232}Th and their decay chain nuclides. The geometric detector area of 144 cm² was realized by nine Cadmium Zinc Telluride (CZT) arrays, each 4 cm x 4 cm (5 mm thick), composed of 256 (16x16) pixels (size: 2.5 mm x 2.5 mm). Each CZT array was readout using two closely mounted Application Specific Integrated Circuits (ASICs), which provided self-triggering capability.

The detector was biased at the cathode with -550 V and the electronic charge signals were collected at the anode. A Cesium Iodide (CsI (TI)) scintillator crystal coupled to photomultiplier tubes (PMT), was used as the anticoincidence system (ACS).

A specially designed collimator provided a field of view (FOV) of $33\text{ km} \times 33\text{ km}$ at the lunar surface from a 100 km orbit. The spatial resolution of HEX was 33 km and the mass was 14.4 kg .

e. Moon Impact Probe(MIP)

The impact probe of 35 kg mass was attached at the top deck of the main orbiter and released during the final $100\text{ km} \times 100\text{ km}$ orbit at a predetermined time to impact at a pre-selected location. During the descent phase, it was spin-stabilized. The total flying time from release to impact on Moon was around 25 minutes.

The primary objective was to demonstrate the technologies required for landing the probe at a desired location on the Moon and to qualify some of the technologies related to future soft landing missions.

There were three instruments on the Moon Impact Probe

- i. Radar Altimeter – for measurement of altitude of the Moon Impact Probe and for qualifying technologies for future landing missions. The operating frequency band is $4.3\text{ GHz} \pm 100\text{ MHz}$
- ii. Video Imaging System – for acquiring images of the surface of the Moon during the descent at a close range. The video imaging system consists of analog CCD camera
- iii. Mass Spectrometer – for measuring the constituents of tenuous lunar atmosphere during descent. This instrument was a commercially available Quadrupole mass spectrometer with a mass resolution of 0.5 amu and sensitivities to partial pressure of the order of $10\text{-}14\text{ torr}$.

The dimension of the impact probe is $375\text{ mm} \times 375\text{ mm} \times 470\text{ mm}$.

f. Chandrayaan-1 X-ray Spectrometer (C1XS)

The primary goal of the C1XS instrument was to carry out high quality X-ray spectroscopic mapping of the Moon, in order to constrain solutions to key questions on the origin and evolution of the Moon. C1XS used X-ray fluorescence spectrometry ($1.0\text{-}10\text{ keV}$) to measure the elemental abundance, and map the distribution, of the three main rock forming elements: Mg, Al and Si. During periods of enhanced solar activity (solar flares) events, it determined the abundance of minor elements

such as Ca, Ti and Fe on the surface of the Moon. The instrument utilised technologically innovative Swept Charge Device (SCD) X-ray sensors, which were mounted behind low profile gold/copper collimators and aluminium/polycarbonate thin film filters. The system had the virtue of providing superior X-ray detection, spectroscopic and spatial measurement capabilities, while also operating at near room temperature. A deployable proton shield protected the SCDs during passages through the Earth's radiation belts, and from major particle events in the lunar orbit. In order to record the incident solar X-ray flux at the Moon, which was needed to derive absolute lunar elemental surface abundances, C1XS also included an X-ray Solar Monitor. With its wide field-of-view of $\pm 52\text{ degrees}$, XSM provides observation of the solar X-ray spectrum from $1\text{-}20\text{ keV}$ with good energy resolution ($< 250\text{ keV}$ at 5 keV) and fast spectral sampling at 16 s intervals. The total mass of C1XS and XSM was 5.2 kg



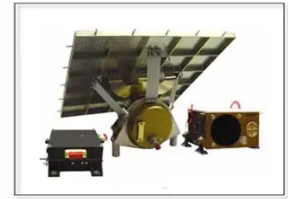
The C1XS instrument is primarily based on the D-CIXS instrument on the ESA SMART-1 mission. Chandrayaan-1 X-ray Spectrometer (C1XS) is realised primarily through ESA funds with partial support from ISRO. The hardware has been developed at the Rutherford Appleton Laboratory, UK

in collaboration with the ISRO Satellite Centre, Bangalore and exhibits significant improvements over the instrument flown on SMART-1.

g. Near Infrared Spectrometer (SIR - 2)

The objective of SIR-2 was to address the surface-related aspects of lunar science in the following broad categories:

- Analyse the lunar surface in various geological/mineralogical and topographical units;
- Study the vertical variation in composition of crust
- Investigate the process of basin, maria and crater formation on the Moon;
- Explore “Space Weathering” processes of the lunar surface
- Survey mineral lunar resources for future landing sites and exploration



SIR-2 was a grating NIR point spectrometer working in the 0.93-2.4 microns wavelength range with 6 nm spectral resolution. It collected the Sun's light reflected by the Moon with the help of a main and a secondary mirror. This light is fed through an optical fiber to the instrument's sensor head, where it was reflected off a dispersion grating. The dispersed light reaches a detector, which consisting of a row of photosensitive pixels that measured the intensity as a function of wavelength and produces an electronic signal, which is read out and processed by the experiment's electronics. The mass of the instrument was 3.3 kg and the instrument unit dimension is 260 mm x 171 mm x 143 mm.

SIR-2 is developed by the Max-Planck-Institute for Solar System Science, through the Max-Planck Society, Germany and ESA.

h. Sub keV Atom Reflecting Analyzer (SARA)

SARA imaged the Moon surface using low energy neutral atoms as diagnostics in the energy range 10 eV - 3.2 keV to address the following scientific objectives:

- Imaging the Moon's surface composition including the permanently shadowed areas and volatile rich areas
- Imaging the solar wind-surface interaction
- Imaging the lunar surface magnetic anomalies

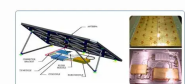
The SARA instrument consisted of neutral atom sensor CENA (Chandrayaan-1 Energetic Neutrals Analyzer), solar wind monitor SWIM and DPU (Data Processing Unit). CENA and SWIM interface with DPU, which in turn interfaces with the spacecraft. The masses of CENA, SWIM and DPU



are 2 kg, 0.5 kg and 2 kg respectively, totaling the SARA mass as 4.5 kg. SARA is realised through ESA, in collaboration with Swedish Institute of Space Physics, Sweden and Space Physics Laboratory, Vikram Sarabhai Space Centre, ISRO. The Data Processing Unit of this payload/experiment is designed and developed by ISRO, while Swedish Institute of Space Physics develops the payload.

i. Miniature Synthetic Aperture Radar (Mini SAR)

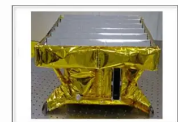
The objective of Mini-SAR was to detect water ice in the permanently shadowed regions on the Lunar poles, upto a depth of a few meters. An onboard SAR at suitable incidence would allow viewing of all permanently shadowed areas on the Moon, regardless of whether sunlight is available or the



angle is not satisfactory. The radar would observe these areas at incidence angle near 45 degrees, recording echoes in both orthogonal senses of received polarization, allowing ice to be optimally distinguished from dry lunar surface. The Mini-SAR radar system operated as an altimeter/scatterometer, radiometer, and as a synthetic aperture radar imager. The Mini-SAR system transmitted Right Circular Polarization (RCP) and received, both Left Circular polarization (LCP) and RCP. In scatterometer mode, the system measured the RCP and LCP response in the altimetry footprint, along the nadir ground track. In radiometer mode, the system measured the surface RF emissivity, allowing determination of the near normal incidence Fresnel reflectivity. Meter-scale surface roughness and circular polarization ratio (CPR) were determined for this footprint. This allowed the characterization of the radar and physical properties of the lunar surface (e.g., dielectric constant, porosity) for a network of points. When directed off nadir, the radar system will image a swath parallel to the orbital track by delay/Doppler methods (SAR mode) in both RCP and LCP. The synthetic aperture radar system worked at a frequency 2.38 GHz, with a resolution of 75 m per pixel from 100 km orbit and its mass was 8.77 kg. Miniature Synthetic Aperture Radar (MiniSAR) is from Applied Physics Laboratory, Johns Hopkins University and Naval Air Warfare Centre, USA through NASA.

j. Moon Mineralogy Mapper (M3)

The primary Science goal of M3 was to characterize and map lunar surface mineralogy in the context of lunar geologic evolution. This translates into several sub-topics relating to understanding the highland crust, basaltic volcanism, impact craters, and potential volatiles. The primary exploration goal was to assess and map lunar mineral resources at high spatial resolution to support planning for future, targeted missions. The M3 scientific instrument was a high throughput pushbroom imaging spectrometer, operating in 0.7 to 3.0 μm range. It measures solar reflected energy, using a two-dimensional HgCdTe detector array.



Sampling: 10 nanometers

Spatial resolution: 70 m/pixel [from 100 km orbit]

Field of View: 40 km [from 100 km orbit]

Mass: 8.2 kg

The spectral range 0.7 to 2.6 μm captures the absorption bands for the most important lunar minerals. In addition, the spectral range 2.5 to 3.0 μm is critical for detection of possible volatiles near the lunar poles. The presence of small amounts of OH or H₂O can be unambiguously identified from fundamental absorptions that occur near 3000 nm.

M3 measurements are obtained for 640 cross track spatial elements and 261 spectral elements. This translates to 70 m/pixel spatial resolution and 10 nm spectral resolution (continuous) from a nominal 100 km polar orbit for Chandrayaan-1. The M3 FOV is 40 km in order to allow contiguous orbit-to-orbit measurements at the equator that will minimize lighting condition variations. Moon Mineralogy Mapper (M3) payload is from Brown University and Jet Propulsion Laboratory, USA through NASA

k. Radiation Dose Monitor (RADOM)

RADOM aimed to qualitatively and quantitatively characterise the radiation environment in near lunar space, in terms of particle flux, dose rate and deposited energy spectrum. The specific objectives are to



- Measure the particle flux, deposited energy spectrum, accumulated radiation dose rates in Lunar orbit;

- Provide an estimate of the radiation dose around the Moon at different altitudes and latitudes;
- Study the radiation hazards during the Moon exploration. Data obtained will be used for the evaluation of the radiation environment and the radiation shielding requirements of future manned Moon missions.

Radiation exposure of crew members on future manned space flight had been recognised as an important factor for the planning and designing of such missions. Indeed, the effects of ionising radiation on crew health, performance and life expectancy are a limitation to the duration of man's sojourn in space. Predicting the effects of radiation on humans during a long-duration space mission requires i) accurate knowledge and modelling of the space radiation environment, ii) calculation of primary and secondary particle transport through shielding materials and through the human body, and iii) assessment of the biological effects of the dose. The general purpose of RADOM is to study the radiation hazards during the Moon exploration. Data obtained will be used for the evaluation of radiation environment and radiation shielding requirements for future manned lunar missions. RADOM was a miniature spectrometer-dosimeter containing one semiconductor detector of 0.3 mm thickness, one charge-sensitive preamplifier and two micro controllers. The detector weighs 139.8 mg. Pulse analysis technique is used for obtaining the deposited energy spectrum, which is further converted to the deposited dose and flux in the silicon detector. The exposure time for one spectrum is fixed at 30 s. The RADOM spectrometer measured the spectrum of the deposited energy from primary and secondary particles in 256 channels. RADOM mass was 160 g. RADOM is from Bulgarian Academy of Sciences.

Salient science results from Chandrayaan-1

1. Discovered the presence of Hydroxyl & water molecules on the lunar surface (M3), lunar exosphere (ChACE on MIP) and sub surface water-ice deposits in the base of the craters of permanent sun shadow region (Mini-SAR)
2. Direct evidence for water (H₂O) in the sunlit ambience from CHACE on MIP of Chandrayaan-1.
3. Reflection of solar wind protons as neutral hydrogen from lunar surface (SARA)
4. Three dimensional conceptualization of many craters of interest and detailed maps of lunar surface features (TMC and LLRI).
5. Detection of potential site (buried lava tube) for future human habitability on the Moon (TMC and HySI). This may provide a safe environment from hazardous radiations, micro-meteoritic impacts, extreme temperatures and dust storms.
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7. Evidences of volcanic vent, lava pond and lava channels as recent as 100 million years old inside the Tycho crater central peak (TMC and NASA's LRO data).
8. Enhanced signatures of OH/ Water in Compton Belkovich Volcanic Complex, which indicates the endogenic water inherited during the formation of the Moon, now coming out with Volcanism (M3 spectra).